

110 - 26/03/2024

In the past laboratory sessions, we have addressed the problem of denoising by going to use the orthogonal matching pursuit algorithm. In that case, we used a dictionary D that was learnt from patches of a natural images. In this lecture we'll see how to actually learn the dictionaries.

Let's define our starting model:

1. $S = DX$
2. $S \in \mathbb{R}^H$
3. $D \in \mathbb{R}^{H \times N}$, $N > H$
4. $X \in \mathbb{R}^N$, X sparse

We have already seen a way to learn a dictionary D by using PCA. In this case, the best case was an orthonormal matrix D . The premises of our new method are the same of the one with PCA. So:

1. INPUT: training set $S \in \mathbb{R}^{H \times W}$. Each column is a vector representing a signal or a patch
2. GOAL: learn D by solving the problem: $\hat{D}, \hat{X} = \underset{\substack{D \in \mathbb{R}^{H \times N} \\ X \in \mathbb{R}^{N \times W}}}{\operatorname{argmin}} \|S - DX\|_F^2$. X is the matrix containing the coefficient vector for each signal in S
 $\hookrightarrow \sum_{i=1}^W \|S[:, i] - D \cdot X[:, i]\|_2^2 \quad \forall i: X[:, i] \text{ is sparse}$

The latter problem can be solved using the K-SVD algorithm. Let's now see the relevant steps of this algorithm:

- 1) INITIALIZE D , where $D \sim N(0, I^2)$ or we can put in D the first N column of S
- 2) ZERO MEAN D : we remove the mean value from each column of $D \rightarrow D[:, i] = D[:, i] - \operatorname{mean}(D[:, i]) \quad \forall i = 1, \dots, N$
- 3) NORMALIZE COLUMNS OF D : $D[:, i] = D[:, i] / \|D[:, i]\|_2$
- 4) SET TO ZERO MEAN S : $S[:, i] = S[:, i] - \operatorname{mean}(S[:, i]) \quad i = 1, \dots, N$

K-SVD steps:

- 5) FIXING D : $\forall i = 1, \dots, W \rightarrow X[:, i] = \underset{X}{\operatorname{argmin}} \|S[:, i] - DX\|_2^2$ where X is sparse
- 6) FIXING X : $D = \underset{D}{\operatorname{argmin}} \|S - DX\|_F^2$

for $l = 1, \dots, \text{max_iteration}$

SPARSE CODING

for $i = 1, \dots, W$

$X[:, i] = \text{OMP}(S[:, i], D, L, \tau = 0)$

If we set L small and $\tau = 0$ then the OMP is going to be very fast

- $\rightarrow L$ small
- $\rightarrow \tau = 0$
- $\rightarrow \text{max iteration} = 10$
- $\rightarrow W = 10000$

for $j_0 = 1, \dots, N$

$$E = S - \sum_{\substack{j=j_0 \\ j \neq j_0 \\ i=1}}^N D[:, j] X[j, i]$$

$$W = \{i = 1, \dots, W : X[j_0, i] \neq 0\}$$

if W is empty

then replace $D[:, j_0]$

$$D[:, j_0] \sim N(0, I^2)$$

or

$$D[:, j_0] = S[:, n] \quad \text{where } n \text{ is a random number between } 1 \text{ and } W$$

else

zero-mean, normalize $D[:, j_0]$

$$\hat{D}, \hat{J} = \underset{\substack{D \in \mathbb{R}^{H \times N} \\ J \in \mathbb{R}^W}}{\operatorname{argmin}} \|E_W - DJ^T\|_F^2$$

$$E_w = U \bar{\Sigma} V^T \quad \text{where SVD: } \bar{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \ddots \end{bmatrix}$$

$$\hat{d} = U[:, 1]$$

$$\hat{g} = \sigma_1 \cdot V[:, 1]$$

$$D[:, j_0] = \hat{d}$$

$$X[j_0, w] = \hat{g}$$

Let's now consider the INPANTING procedure. In this case, our signal s is something like $s = P(Dx + \eta)$, where the projection P basically removes some pixels. Even though we have a dictionary D learnt from the K-SVD,

it is not enough to reconstruct the signal s because we need to add a constant atom. So, considering our model:

$s = P(Dx + \eta)$, we have that: $\text{mean}(s) \neq \text{mean}(Dx)$. We can use the OMP to estimate both the non-constant

part of s and the average value of γ . Hence, $\hat{x} = \text{OMP}(s, P(D), L, \tau)$ and $s = D\hat{x}$. We need to allow our

dictionary to reconstruct also the average value: $D_{\text{extended}} = [D, \underline{c}]$ where $\underline{c} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \cdot \frac{1}{\sqrt{M}}$ and $\|\underline{c}\|_2 = 1$

Therefore, $\text{mean}(\hat{s}) = \hat{x}[-1]$